



2. Strong interaction



STARTSEITE

EINFÜGEN

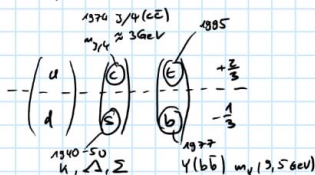
ZEICHNEN

ANSICHT



Beenden

2) Quarks and colour

Family of quarksspin- $\frac{1}{2}$ particlesmesons $|q \bar{q}'\rangle$ $|\bar{q} q'\rangle$ different quarks
of which at a different timebaryons $|q q' q''\rangle$ $|\bar{q} \bar{q}' \bar{q}''\rangle$ colour

all up quarks in spin-up state

1951 Fermi $\Delta^{++} = |u u u\rangle$ $J_\Delta = \frac{3}{2}$

forbidden by Fermi statistics (quantum numbers have to differ)

 $\Rightarrow$ we need another quantum number: colour charge3 colours R, G, B and 3 anti colours $\bar{R}, \bar{G}, \bar{B}$

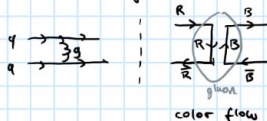
hadrons are 'colourless'

 $|R \bar{R}\rangle, |G \bar{G}\rangle, |B \bar{B}\rangle$ $|R G B\rangle$ + combinationsgluons

- mediator of the strong interaction

 $m_0 = 0$

- bicolour objects



- colour-anticolour state

3 possibilities





2. Strong interaction



STARTSEITE

EINFÜGEN

ZEICHNEN

ANSICHT



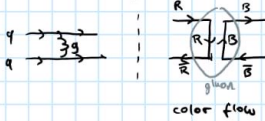
Beenden

gluons

- Mediator of the strong interaction

$$m_0 = 0$$

- bicolour objects



- colour-anticolour state

3 possibilities

but 1 is colourless

$$|R\bar{R}\rangle + |B\bar{B}\rangle + |G\bar{G}\rangle \Rightarrow \text{gluon does not exist}$$

$$\Rightarrow \underline{8 \text{ gluons}}$$
next time: $SU(2) = \text{isospin}$ $SU(3) = \text{colour, low-mass quarks and strong model}$



2. Strong interaction



STARTSEITE

EINFÜGEN

ZEICHNEN

ANSICHT



Beenden

1.5) Deep inelastic scattering

the γ interacts with one constituent of the proton

→ historically called partons (Feynman)



for high-energy scattering

$$E \gg m_p$$

the struck parton carries a fraction of the total momentum

of the proton

$$x = \frac{Q^2}{2p \cdot q} \quad 0 \leq x \leq 1$$

$\uparrow$ fraction of the proton
 $\uparrow$ momentum of the γ

parton distribution function (PDF)

for the up-quark PDF

$$u(x) \quad S(x)$$

$\uparrow$
 determined experimentally

⇒ number of up-quarks that have a momentum fraction between

 x and $x + \delta x$

PDF cannot be computed from theoretical calculation

⇒ non-perturbative regime

⇒ PDF are determined experimentally

electron-target exp
 pp collider LHC
 pp̄ collider Tevatron

very high energy colliders

$$\sqrt{s} \gg 1$$

the fraction of partons with low x increases

Illustration





Beenden

Strong interaction

Mittwoch, 9. September 2026

09:38

09.09.26

- Deep inelastic scattering (experimental)
- Quark model (theoretic)
- Symmetries (in the quark model) (theoretic)
- Hadronization (experimental)

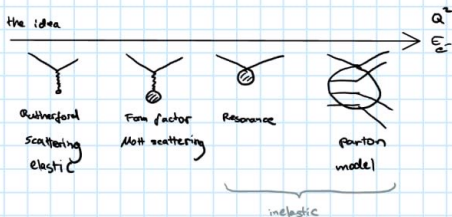
1) Deep inelastic scattering

1.1) Probing the structure of the proton



We define $Q^2 = -q^2 > 0$ for t-channel processes

photon wavelength $\lambda \approx \frac{1}{\sqrt{Q^2}}$



1.2) Inelastic scattering at low Q^2

scattering when the final state contains excited states



for example $e^- p \rightarrow e^- \Delta^+ \rightarrow e^- p \pi^0$

1.3) Deep inelastic scattering

the γ interacts with one constituent of the proton





2. Strong interaction



STARTSEITE

EINFÜGEN

ZEICHNEN

ANSICHT



Beenden

Illustration

Tevatron and LHC collider

$$pp \rightarrow t\bar{t} \text{ LHC}$$

very low mass

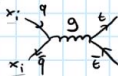
$$p\bar{p} \rightarrow t\bar{t} \text{ Tevatron}$$

threshold production

$$x \approx \frac{2m_b}{\sqrt{s}}$$

fraction at momentum

Tevatron



fraction of momentum

LHC



②

each parton has to have at least x

minimum value of x to produce a pair of quarks

$$\text{Tevatron } \sqrt{s} = 1.96 \text{ TeV}$$

$$m_t \approx 175 \text{ GeV}$$

$$x \approx 0.175$$

$$\text{LHC } \sqrt{s} \approx 14 \text{ TeV}$$

$$x \approx 0.025$$

 $\Rightarrow$ it's enough for a parton to have 2.5% of the momentum of the proton

proton

high v

low v

p

u

u

d

valence quarks

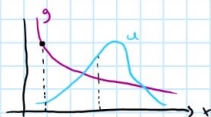
- quarks with highest probability to be seen



sea quarks

 e^-p  $e^-p \rightarrow \gamma^* S$

PDF (gluon)

 $\rightarrow$ explanation why at LHC ② is more likely than ①

2) Quarks and colour

Family of quarks

$$1370 \text{ } 3/4 (c\bar{c})$$

$$m_{c\bar{c}} \approx 3 \text{ GeV}$$

885

